Triadic Casimir — PKLSA v0.3.0 Framework Report

SST Workbench / Unregistered instance

October 14, 2026

1 Instance identity

A065-v0.3.0 versioned Triadic-Casimir research instance.

2 Research question

Does a preregistered triadic clock/stress geometry pilot satisfy its analytic controls while independent physical coupling gates remain unresolved?

3 Null and alternative hypotheses

No independently qualified source-specific physical two-surface response is established by the present geometry and toy models.

4 Explicit assumptions

Time-dependent regularized filament RK4 is exploratory and is not full Euler closure.

5 Definitions

u material velocity; omega=curl u; S symmetric strain.

6 Analytical equations

The official AUTO_SCIENCE_CONTRACT fragment records frozen equations.

Machine-readable science contract

Research question Does a preregistered triadic clock/stress geometry pilot satisfy its analytic controls while independent physical coupling gates remain unresolved?

Objective Independently verify three dataset seals, analytic Euler/gauge/sheet controls and one sourced KnotPlot trefoil finite-core geometric response without importing target force.

H_0 No independently qualified source-specific physical two-surface response is established by the present geometry and toy models.

H_1 Independently qualified PKLSA Euler motion, boundary conditions and microscopic link coupling produce held-out physical predictions.

Registered equations

E1: Incompressible Euler kinematics

$$\nabla \cdot \mathbf{u} = 0, \quad \boldsymbol{\omega} = \nabla \times \mathbf{u} \tag{1}$$

Dimensional check: divergence and vorticity both inverse seconds

E2: Pressure Poisson source

$$\nabla^2 p = \rho_M(\frac{1}{2}|\omega|^2 - ||S||_F^2) \tag{2}$$

Dimensional check: kg/(mˆ3 sˆ2) on each side

E3: Regularized geometric velocity diagnostic

$$\mathbf{u}(\mathbf{x}) = \frac{1}{4\pi}\sum_j \frac{\Delta\mathbf{X}_j \times (\mathbf{x} - \mathbf{X}_{j+1/2})}{(|\mathbf{x} - \mathbf{X}_{j+1/2}|^2 + \varepsilon^2)^{3/2}} \tag{3}$$

Dimensional check: dimensionless Gamma=1 geometry diagnostic

Registered code-to-mathematics steps

1. **S1** [G1; equations E1]: Verify all three source seals before any scientific result is evaluated
2. **S2** [G2; equations E2, E3]: Evaluate analytic Euler controls and preregistered trefoil geometry resolution scan
3. **S3** [G3; equations E3]: Withhold physical Casimir claims without independent providers, wall dynamics, and atom-link normalization

7 Scientific algorithm

Verify three seals; compute controls; evaluate trefoil diagnostic.

8 Dataset provenance

D01 D02 D03, plus one KnotPlot relaxed N300 trefoil in D01.

9 Independent gates

Only source/analytic gates qualify; dynamic physical claims remain unresolved.

10 Numerical controls

Dimensionless finite-segment Biot-Savart and finite-difference derivatives.

11 Backend authority

CPU profile enables strict C++ gate; prerequisite G3 and native physical validation remain unresolved.

12 Blindness and integrity

All three dataset SHA256 seals checked; reveal comparator not used in blind kernel.

13 Statistics

Geometry subsamples are not independent upstream providers.

14 Result status

Read dynamic AUTO_RESULTS fragment; no target-matched force.

Automatic run provenance

Field	Recorded value
Protocol bundle SHA-256	93d15ecd6109b50781e3fd18d2118e2833fc6ed18c183a9b68bee00d53e8d0c0
Framework schema	SST-FALSIFIER-PROTOCOL-2
Python	3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)]
Platform	Windows-11-10.0.26300-SP0
Output manifest SHA-256	afa76147f3411357dfa6984b8493451d71290d66b4d4829edd439897f5d12b52

Gate ledger

Gate	Status	Question
G0	PASS	Is the frozen protocol intact and is the runtime self-consistent?
G1	PASS	Are all preregistered sources present, identified and provenance-qualified?
G2	FAIL	Do reference numerics satisfy analytic/sanity and dimensional checks?
G3	NOT_RUN_PREREQUISITE	Does target-free discovery reject the preregistered null under the registered statistic
G4	NOT_RUN_PREREQUISITE	Does an independent held-out/provenance-family confirmation reproduce the discov
G5	NOT_RUN_PREREQUISITE	Does strict native CPU FP64 certification reproduce the authoritative reference wit
G6	NOT_APPLICABLE	If enabled, does GPU screening pass CPU to GPU parity without being promoted b
G7	NOT_APPLICABLE	Does the result replicate across preregistered source families/encodings without pseu
G8	NOT_RUN_PREREQUISITE	Does the preregistered physical/mechanism provider satisfy its convergence and mec
G9	DEFERRED	After all blind gates, does reveal verification match the preregistered commitment a

Backend manifest


```
{
  "note": "Reference geometry plus actual regularized filament ODE; CPU profile selected but physical G3
  "physical_force_status": "UNRESOLVED",
  "pklsa_geometry": "a60b1a95a00110e3cb32087e7b4328d16e967379ebf97d60117b0ec05692ffb2",
  "python_reference": {
    "actual_backend": "python-stdlib-binary64",
    "authority": "REFERENCE",
    "precision": "float64"
  },
  "schema": "SST-BACKEND-MANIFEST-2",
  "sealed_source_groups": [
    "D01_EULER",
    "D02_CLOCK_LINK",
    "D03_TWO_SURFACE"
  ]
}
```


Source manifest


```
{
  "manifest_sha256": "2eac49e6321ba11a21c38691afee55c5f712982bde0c9996d7797708114121b8",
  "schema": "SST-SOURCE-MANIFEST-2",
  "sources": [
    {
      "catalog_id": "A046",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "synthetic-euler",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
      "provenance_family": "synthetic-euler",
      "registry_hint": {
        "catalog_id": "A046",
        "hits": []
```

```
      },
      "role": "analytic_control",
      "sha256": "9c4da362ab58a546fe8aff349a024dcc7fb63218ef30053dceee3d6659c574ad",
      "source_id": "D01",
      "version": "v0.3.0"
    },
    {
      "catalog_id": "E010",
      "evidence_class": "simulation",
      "exists": true,
      "independence_family": "knotplot-relaxed-3_1",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
      "provenance_family": "knotplot-relaxed-3_1",
      "registry_hint": {
        "catalog_id": "E010",
        "hits": []
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      "role": "geometry_diagnostic",
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      "source_id": "PKLSA",
      "version": "v0.3.0"
    },
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      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "synthetic-link",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
      "provenance_family": "synthetic-link",
      "registry_hint": {
        "catalog_id": "CANON",
        "hits": []
      },
      "role": "toy_control",
      "sha256": "8e4bcfcdf186e539f0cdeafd9cbbf9f31dbe7bbfb95c14d89099e6c31f760af8",
      "source_id": "D02",
      "version": "v0.3.0"
    },
    {
      "catalog_id": "CANON",
      "evidence_class": "synthetic_control",
      "exists": true,
      "independence_family": "synthetic-sheet",
      "path": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Casimir_B
      "provenance_family": "synthetic-sheet",
      "registry_hint": {
        "catalog_id": "CANON",
        "hits": []
      },
      "role": "heldout_control",
      "sha256": "fff8860c2597840d0f15906a43f8710793c0909ebfdde1b1eb6f64aea7863adc",
      "source_id": "D03",
      "version": "v0.3.0"
    }
  ],
  "workbench_root": "C:\\workspace\\projects\\SST-Workbench\\01_research\\A_falsifiers\\A065_Triadic_Cas
}
```


15 Scientific interpretation

Physical Casimir force and gravitation remain unvalidated.

16 Reproduction

Run native FREEZE and FULL via the installation bridge.